6 Lecture 6 (January 24, 2025): Symmetric Braidings and Rigidity

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6.1 Definition. A braiding $c_{X,Y} : X \otimes Y \to Y \otimes X$ is *symmetric* if

$$c_{Y,X} c_{X,Y} = \mathrm{Id}_{X \otimes Y}.$$

6.2 Remark.

- Let $\mathcal{C}$ be braided and take $X \in \mathcal{C}$. Then we get an action of the braid group

$$B_n = \langle b_1, \ldots, b_{n-1} \mid \begin{array}{ll} b_i b_j = b_j b_i & (|i-j|>1), \\ b_i b_{i+1} b_i = b_{i+1} b_i b_{i+1} & (1 \le i \le n-2) \end{array} \rangle$$

  on $X^{\otimes n}$, with b_i acting by $c_{X,X}$ in positions $i, i+1$ and identities elsewhere. Coherence supplies the required associators.

- If the braiding is symmetric, this action factors through $S_n = B_n / \langle b_i^2 \rangle$

6.3 Example. The "swap" braiding in $\mathbb{V}\mathrm{ec}_k$

$$x \otimes y \mapsto y \otimes x$$

is symmetric.

6.4 Example. For every affine group scheme G, the ordinary flip is G-equivariant for the diagonal action, so its representation category is symmetric as well.

6.5 Example. Take an abstract group G. Define the category $\mathbb{V}\mathrm{ec}_k\, G$ of G-graded k-vector spaces whose objects are vector spaces of the form

$$V = \bigoplus_{g \in G} V_g$$

and whose morphisms are degree-preserving linear maps.
For any G, this has a monoidal structure given by

$$(V \otimes W)_g = \bigoplus_{xy=g} (V_x \otimes W_y).$$

In this case, a braided structure exists if and only if G is abelian. To see the only if direction, notice that for any $g \in G$ we have an object δ_g which is k in degree g and 0 in all other degrees. If G has a braiding, then we get

$$\delta_{gh} = \delta_g \otimes \delta_h \simeq \delta_h \otimes \delta_g = \delta_{hg},$$

and so $gh = hg$. Conversely, for abelian G, the ordinary flip preserves the grading and satisfies the hexagon identities for the usual associator.

6.6 Example. Consider $\mathrm{SVec}_k = \mathbb{V}\mathrm{ec}_k(\mathbb{Z}/2\mathbb{Z})$, vector spaces graded by $\mathbb{Z}/2\mathbb{Z}$. In particular, every object is of the form $V = V_0 \oplus V_1$, where we call V_0 and V_1 the even and odd homogeneous components, respectively.
For $v \in V_i$, we define $|v| = \deg v = i$. Then the braiding $V \otimes W \to W \otimes V$ is given on homogeneous elements by

$$v \otimes w \mapsto (-1)^{|v||w|} w \otimes v.$$

The sign is multiplicative in each degree, so the hexagon identities hold; applying the flip twice gives sign $(-1)^{2|v||w|} = 1$, hence symmetry. When $\operatorname{char} k = 2$, the sign equals 1 and this braiding is the ordinary flip on graded spaces. The distinction considered below requires characteristic different from 2.

6.7 Definition. Fix a symmetric monoidal category $\mathcal{C}$ and $X \in \mathcal{C}$. A *dual object* of X is an object X^* with maps $\mathrm{ev}_X : X^* \otimes X \to \mathbb{1}$ and $\mathrm{coev}_X : \mathbb{1} \to X \otimes X^*$ such that

$$(\mathrm{id}_X \otimes \mathrm{ev}_X) a_{X,X^*,X} (\mathrm{coev}_X \otimes \mathrm{id}_X) = \mathrm{id}_X,$$
$$(\mathrm{ev}_X \otimes \mathrm{id}_{X^*}) a^{-1}_{X^*,X,X^*} (\mathrm{id}_{X^*} \otimes \mathrm{coev}_X) = \mathrm{id}_{X^*},$$

where the canonical unit constraints identify the sources and targets with X and X^*.

6.8 Definition. $\mathcal{C}$ is *rigid* if all objects admit a dual.

6.9 Remark.

- If X^* exists, it is unique up to a unique isomorphism compatible with evaluation and coevaluation.
- Rigidity is a property, not a structure.
- If $\mathcal{C}$ is rigid, we get a contravariant functor $X \mapsto X^*$.
- In a symmetric category, we get a natural isomorphism $X \cong (X^*)^*$.
- Rigidity can be defined for any monoidal category. Without the symmetric braiding, we get "left" and "right" duals X^* and ${}^*X$, respectively, such that $X \cong {}^*(X^*)$.

6.10 Example. In $\mathbb{V}\mathrm{ec}_k$, we can let $V^* \coloneqq \mathrm{Hom}_k(V, k)$ with

$$\begin{aligned} \mathrm{ev}_V : V^* \otimes V &\to k \\ f \otimes v &\mapsto f(v) \end{aligned}$$

and

$$\begin{aligned} \mathrm{coev}_V : k &\to V \otimes V^* \\ 1 &\mapsto \sum_i v_i \otimes f_i, \end{aligned}$$

where $\{v_i\}$ is a finite basis and $\{f_i\}$ its dual basis. A coevaluation satisfying the triangle identities exists precisely when V is finite dimensional. The tensor

$\sum_i v_i \otimes f_i$ corresponds to id_V under $V \otimes V^* \simeq \mathrm{End}_k(V)$, so it is independent of the basis. Conversely, if a coevaluation satisfying the triangle identities has $\mathrm{coev}(1) = \sum_{i=1}^m v_i \otimes f_i$, the first identity gives $v = \sum_i f_i(v) v_i$ for every $v \in V$. Thus the v_i span and V is finite dimensional.

6.11 Definition. An abelian k-linear category $\mathcal{C}$ is *Artinian* if

1) Hom sets are finite dimensional k-vector spaces
2) All objects have finite length. That is, for any $X \in \mathcal{C}$ there is a family of strict inclusions
$$0 = X_0 \subset X_1 \subset \cdots \subset X_n = X$$
such that each X_i/X_{i-1} is simple.

6.12 Definition. A *symmetric tensor category* is a category $\mathcal{C}$ which has

1) A rigid symmetric monoidal structure.
2) An Artinian abelian k-linear structure.

such that

1) $\otimes$ is k-linear on Homs (that is, bilinear).
2) $\mathrm{End}_{\mathcal{C}}(\mathbb{1}) \cong k$. Under these hypotheses the unit is simple; see [EGNO15, Section 4.3].

6.13 Remark. Rigidity implies that tensoring with an object has both a left and a right adjoint, given by tensoring with suitable duals. A functor between abelian categories with both adjoints is exact. Hence $\otimes$ is exact in each variable; bilinearity alone would not imply this. See [EGNO15, Proposition 4.2.1].

6.14 Exercise. Show that G-graded vector spaces are equivalent to right comodules over the group algebra kG, by sending $v \in V_g$ to $v \otimes g$.

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